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Isolation and identification of pathogenic bacteria and its antibacterial susceptibility analysis in edible fish *Catla catla*

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Abstract Fishes are frequently affected by bacteria and virus. Bacterial diseases have become major impediments to aquaculture, especially when water temperature is warm. In the present study, the pathogenic bacteria was isolated from the infected fish *Catla catla.* Identification process was carried out using Biochemical test. MacConkey agar medium was prepared for the confirmation for the species level identification of pathogenic bacteria that were isolated from the infected fish *Catla catla*. Finally, *Serratiamarcescens* species were identified from the intestine of infected fish. Antibacterial discs were treated against the *Serratia*species. Different zone of inhibition was recorded for confirming the sensitive and resistance against the *Serratiamarcescens.* Among the 13 antibacterial discs, the certain antibiotic was sensitive such as Chloramphenicol -30mm, Streptomycin -29mm, Kanamycin -29mm, Gentamicin -26mm, against the pathogenic bacteria *Serratiamarcescens.* Results of the present study showed that Chloramphenicol, Streptomycin, Kanamycin and Gentamicin can be used as a drug against the pathogenic bacteria *S. Marcescens* for the infected fish (*Catla catla*). The above mentioned Antibacterial disc against *S. Marcescens* induced the growth of *Catla catla* and promote the production and increase the Economic status of Aquaculture.

Key Words *Catla catla*, *Serratia marcescens*, MacConkey agar medium, Antibacterial disc.

Introduction

Aquaculture is one of the fastest emerging fields in industrial sectors for producing animal feed which greatly contributes to the world's well-being and prosperity. According to FAO statistics, aquaculture provided nearly 50% of the annual world fisheries production of 120 million metric tons in 2004 (FAO, 2013). Of this, half is comprised of finfish. Allison and Ellis (2001) has described aquaculture is the farming and husbandry of economically important aquatic animals and plants under controlled conditions. Fish have great significance in the life of mankind, being an important natural source of protein and providing certain other valuable products and economic status to several nations. The quality of fish and their spoilage is always been the severe apprehension for human health. Therefore many researchers studied the quality of different fishes associated with micro flora and fauna (Subhash Walke*et. al*., 2015; Alikunhi *et. al*., 2016). Disease outbreaks are being increasingly recognized as a significant constriction on aquaculture where fish are stocked at high intensities and subjected to stress, thus affecting the economic status. Microbes play an important role in affecting

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fish health (Sujatha *et al.*, 2013). The fish pathogen may be grouped as bacteria, fungi, virus etc. These pathogens are responsible for diseases its due to 45% profit reduced in fishfarms (Kabata, 2008). Fish suffer from the various type of disease. All fishes normally carry pathogens and parasite, and if the pathogenic load increases it lead to disease. Diseases act as the primary role in affecting fish mortality, especially when fish are the young stage (Sharma *et al*., 2012).

Bacteria are one of an important causative agent of fish diseases in both wild and cultured fish and are responsible for severe profitable losses (Omprakash and Ravi, 2015). Fish obtain bacteria in the digestive tract from the aquatic environment through water and food particles (Chovatiya *et al.*, 2014). A number of bacteria produce a similar syndrome, generically referred to as hemorrhagic septicemia and characterized by exterior reddening and hemorrhage in the peritoneum, body wall, and viscera. Morbidity and mortality are highly variable, depending on predisposing conditions such as low dissolved oxygen, other water quality problems, handling stress or trauma. Ulcerative lesions are common as a disease progresses, and mortality can be significant if stress is not controlled. Antibiotic therapy is recommended if fish are dying. Common bacterial isolates from affected fish include *Aeromonas* and *Pseudomonas sp.*, which are more common in freshwater animals, and *Vibrio sp.*, more commonly isolated from marine fishes.

The bacterial species belonging to 13 genera have been reported to be pathogenic to aquatic animals or organisms, including (1) Gram-negative bacteria such as *Aeromonas*, *Edwardsiella*, *Flavobacterium*, *Francisella*, *Photobacterium*, *Piscirickettsia*, *Pseudomonas*, *Tenacibaculum*, *Vibrio*, *Yersinia*, *Klebsiella*, *Serratia*. (2) Gram- positive bacteria such as *Lactococcus*, *Renibacterium*, *Streptococcus* .Most bacterial pathogens of fish are aerobic, gram-negative rods. Diseases outbreaks among fishes increase the mortality rate and decrease the productivity leading to the high economic drop to fish farmers (Sharma *et al.*, 2012). To enhance the aquaculture activity one of the best ways is to reduce the disease of fish. Therefore, the aim of the present study is to isolate and identify the Pathogenic bacteria from the Infected Fish (*Catla catla*) and to identify antibiotics for the specific pathogenic bacteria which were isolated from infected *Catla catla*

Objective

1. To isolate and identify the bacterial species in the fish samples which are potentially pathogenic to humans.
2. Toob serve the organ wise (Intestine, Gills, Skin surface) distribution of the bacterial Pathogens.
3. To study the antibacterial susceptibility against the pathogenic bacteria.

Materials and methods

Infected fishes (*Catla catla*) were collected from Aliyar dam, (Figure 1) Coimbatore district, Tamil Nadu, India. It was transported to the laboratory.

Analysis of physico-chemical parameters of water

The physico-chemical parameters viz., temperature, pH, dissolved oxygen, Total alkalinity, hardness, calcium, nitrates, nitrites, ammonia, phosphates, iron, fluorides, chlorides and residual chlorine were estimated by using Indian standard procedure for Aliyar dam water. The Length & weight of infected Fish (*Catla catla*) were measured.

Sample preparation

Preparation of serial dilution and media

The sample was prepared according to Obi and Krakowiaka (1983).The part of the fresh fish sample inserted into the test tube containing 9 ml of distilled water as a stock, and five other test tubes also containing 9 ml of distilled water were arranged serially in the test tube holder. 1 ml of the stock was collected using a pipette to the first test tube and from the first test tube to the second test tube up to the fifth test tube respectively i.e. 10^{-1}, 10^{-2}, 10^{-3}, $10^{-4}$and 10^{-5} respectively. 10^{-4} and 10^{-5} were used as the dilution factor and 1 ml was taken from each factor into a sterilized petridish in duplicate**.** Media was prepared by 2.8 g of Nutrient agar, then autoclaved at a temperature of 121°C for 15 minutes. The media was allowed to cool and solidify.

Culture of bacteria

A reusable glass or metal spreader should be flame sterilized by dipping in alcohol (such as 70% isopropyl or ethanol), shaking off the excess alcohol, and igniting the residue. The spreader is then allowed to cool. The spreader is placed in contact with the inoculum on the surface of the plate and positioned to allow the inoculum to run evenly along the length of the spreader. All plates were incubated at a temperature of 37°C for 24 hrs, before colony counting and isolation procedures.

Bacteria colony count

After 24 hrs incubation, Bacteria colonies were counted using Digital Colony Counter. The number of colonies on the plate was multiplied by the reciprocal of the dilution factor and calculation was done for 1 ml of original sample, and plating was done in duplicate for each dilution. Anaverage calculate was taken to obtain the total count. A pure culture is carried out for further study

Identification of bacteria

Characterization of bacterial isolates

Gram's staining method was carried out to find whether it is gram negative (or) gram positive. Biochemical Test for Bacterial Identification were carried out such as Carbohydrate fermentation (Durhams tube), Indole (Tryptophan degradation) Production test, Methyl red test (mixed fermentation), Voges – Proskauer test (Butanediol fermentation), Citrate utilization test, Catalase test , Oxidase test, Triple sugar iron (TSI) agar test, Urea hydrolysis test (Urease test), Starch hydrolysis test etc..

Antibiotic susceptibility test

Antibiotic susceptibility test was carried out by the disk diffusion technique using a commercially available disc (Hi-media). The antimicrobial sensitivity of the test strains to thirteen antibacterial drugs was done using the Kirby-Bauer disk diffusion method (Bauer et al., 1966). The antibiotics used were ampicillin, chloramphenicol, erythromycin, gentamicin, kanamycin, methicillin, ofloxacin, penicillin G, polymyxin-B, rifampicin, tetracycline, streptomycin and vancomycin.

Results and discussion

In the aquatic environment, fish are in constant interaction with a wide range of pathogenic and non-pathogenic microorganisms (Sujatha *et al.,* 2007). Microbes play an important role in affecting fish health as well as suffer from various types of diseases. The isolation of microorganism is based on the infected fish species, its disease status, clinical signs and biochemical diagnosis.

Aliyar dam water sample was analyzed for various physicochemical parameters such as Temperature, pH, Fluoride, Chloride, Residual Chlorine, Total Hardness, Total Alkalinity, Calcium, Phosphate, Iron, Nitrate, Nitrite, Ammonium and Dissolved Oxygen were measured (Table 1). The Length and Weight of infected Fish Catla (*Catla catla*) were observed (Table 2).

Bacterial colony counts made among the various tissues of infected fish. (Table 3 and Table 4) A comparison of the bacterial count in various organs was analyzed. In our present study, we have been taken the bacterial count from the various organs, such as Intestine, Gills and the Skin surface of the infected *Catla catla*. Among this various organs, the highest amount of bacterial count was observed in Intestine followed by Gills and Skin surface. A number of bacterial pathogens were isolated and identified in freshwater ecosystem since 1970 in India, however, large new emerging strains were developed due to resistant with various environmental and antibiotic resistant mechanism. This is the major reason for the failure of effective protective vaccine or antibiotics against bacterial pathogens. In our knowledge, no effective vaccines were available in our market against bacterial pathogens, however, experimental studies show that formalin, heat killed, lipopolysaccharides provide protective and promising protection against specific pathogens *Serratia marcescens*

The isolated bacteriawas confirmed on the basis of its morphological, Physiological and Biochemical Characteristics as the following Standard Method. In this present study, we took an infected *Catla catla* and isolate the bacteria from the intestine. The result shows that the bacteria was gram negative as well as Rod-shaped (Fig. 2 and Table 5). We have identified that the bacteria present in the intestine was *Serratia marcescens* with the help of the biochemical test and by applying specific medium. In the biochemical test observation, the bacteria was 90 percentage confirmed as *Serratia* genus (Table 5). Similar result was also observed by Camus *et al.*, (2013). In 2012, Shahitha observed the isolated colonies of *Serratia marcescens* and confirmed through the morphology and biochemical characteristics

The MacConkey agar medium was selected for confirming the bacterial species. After 24 hours incubation, a pale white colourcolonies were developed on the medium. It showed the bacteria as Lactose non-fermenter. So we confirmed that the bacteriawas *S.marcescens* (Fig. 3). *Serratia marcescens* is a rod-shaped gram-negative bacterium measuring 0.5-0.8 µm in diameter and 0.9-2.0 µm in length. *S. marcescens* is found on dead organic material, while some inhabit soil, water, air, plants, animals, food particles. *S. marcescens* is identified by their redpigment is called prodigiosin (Kurz *et al.*, 2003). This pigment is easily recognized in certain strains due to its red coloring and it has been extensively used as a biological marker.

Antibacterial susceptibility test was conducted and confirmation of *S.marcescens* was done by using Nutrient agar plates. On the basis of antibacterial resistance, strains were categorized into three different groups. One group had strains like Chloramphenicol (30 mm), Gentamicin (26 mm), Kanamycin (29 mm), Streptomycin (29 mm), Vancomycin (22 mm), Ofloxacin (23 mm), Erythromycin (24 mm) and Tetracycline (24 mm) which were sensitive to the *S. marcescens*. The second group had strain Rifampicin (15 mm) which was moderately susceptible (intermediate) to

the *S. marcescens*. The third group contained strains like Polymixin-B (10 mm), Ampicillin (02 mm), Methicillin (02 mm) and Penicillin G (01 mm) which were resistant to the *S. marcescens* (Table 6 and Fig. 4). Chloramphenicol was highly sensitive when compared than Kanamycin, Streptomycin, Gentamicin was observed. The previous study showed that a multi-resistant characteristic of *S. marcescens* to antimicrobials being sensitive to only 33.3% of the drugs tested. Isolates were resistance to cephalothin, ampicillin and cefoxitin (Saidenberg *et al*., 2007).

Conclusion

From the results, the cultures were identified as *Serratia marcescens*. Chloramphenicol, Streptomycin, Kanamycin and Gentamicin can be used as a drug against the pathogenic bacteria *S. Marcescens* for the infected fish (*Catla catla*) and also induced the growth of *Catla catla*. Thus, it promotes the production and increases the Economic status of aquaculture. These results also enabled us to conclude that there are strains of *Serratia marcescens* which weakly decompose carbohydrates. Thus, it appears more likely to be a water related problem than a fish related one because the bacteria appears to enter the fish through the media. Nevertheless, the presence of a large number of enteric bacteria including pathogenic species in the pond suggests the need for following strict hygienic methods during the process of handling and processing of fish in these systems.

Table 1: Physico-chemical parameters of Aliyar dam water during the sampling day

S. No.	Parameters	Units	Result
1	Atmospheric Temperature	^{o}C	30
2	Water Temperature	^{o}C	24
3	pH	-	8.0
4	Dissolved Oxygen	mg/L	6.8
5	Total alkalinity	mg/L	132
6	Hardness	mg/L	142
7	Calcium	mg/L	30
8	Nitrates	mg/L	5.0
9	Nitrites	mg/L	0.5
10	Ammonium	mg/L	0.58
11	Phosphates	mg/L	0.53
12	Chlorides	mg/L	50
13	Fluoride	mg/L	1
14	Iron	mg/L	0.26
15	Residual Chlorine	mg/L	0.5

Table 2: Length and Weight ranges of the infected fish (*Catla catla*)

S. No	Name	Length	Weight
1	*Catla catla*	31.2cm	503.5 g

Table 3: Bacterial counts in aliyar dam water (per ml)

S. No	Total Bacterial Count at Room Temperature	Total Bacterial Count at 37°C
1.	2.1×10^{4}	1.11×10^{4}

Table 4: Bacterial counts in skin surface, gills and intestine of infected fish (*Catla catla*)

S. No	Tissues	Total Bacterial Count at Room Temperature	Total Bacterial Countat 37°C
1.	Skinsurface	7.1×10^{4}	1.10×10^{5}
2.	Gills	8.6×10^{4}	1.28×10^{5}
3.	Intestine	9.8×10^{4}	1.52×10^{5}

Table 5: Biochemical tests for identification of bacteria

S. No	Test	Result
1	Gram's stain	-ve
2	Morphology	Rod
3	Arrangement	Single
4	Catalase	+ve
5	Oxidase	-ve
6	Indole	-ve
7	MethylRed	-ve
8	Voges-Proskeur	+ve
9	Carbohydrate Fermentation	+ve
10	Citrate Utilization	+ve
11	Triple Sugar Iron Agar (H_2SProduction)	-ve
12	Urea Hydrolysis	-ve
13	Starch Hydrolysis	-ve

Table 6: Antibacterial susceptibility analysis against *Serratia marcescens*

S. No	Antibiotics name		Reaction of Strain	Inhibition Zone (mm)
1.	Ampicillin	AMP	R	02
2.	Chloramphenicol	C	S	30
3.	Erythromycin	E	S	24
4.	Gentamycin	GEN	S	26
5.	Kanamycin	K	S	29
6.	Methicillin	MET	R	02
7.	Ofloxacin	OF	S	23
8.	PenicillinG	P	R	01
9.	Polymyxin-B	PB	R	10
10.	Rifampicin	RIF	I	15
11.	Streptomycin	HLS	S	29
12.	Tetracycline	TE	S	24
13.	Vancomycin	VA	S	22

Fig. 1: Infected fish from aliyar dam

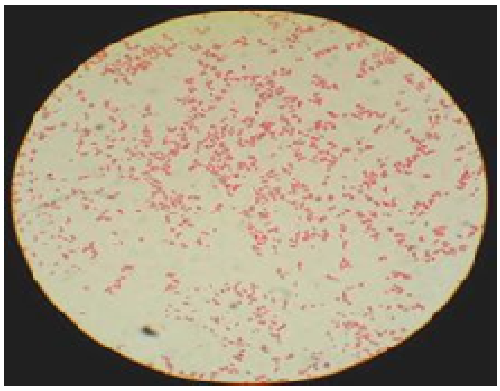

Fig. 2: Gram stain test- gram negative bacteria

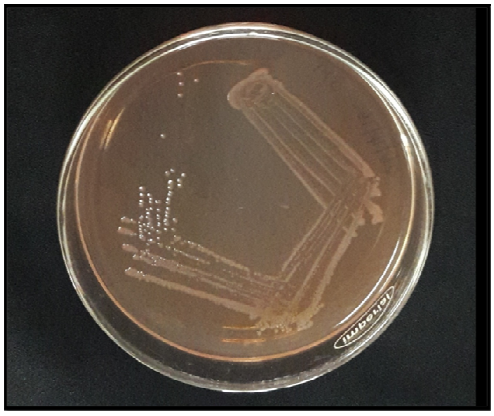

Fig. 3: *Serratia marcescens* on MacConkey agar plate

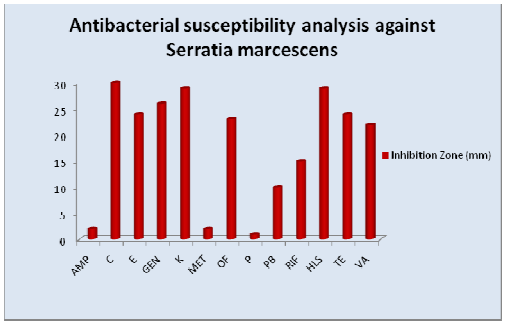

Fig. 4: Antibacterial susceptibility analysis against *Serratia marcescens*

References

Bauer, A.W., Kirby, W.M., Sherris, J.C., Turck, M., 1966. Antibiotic susceptibility testing by a standardized single disk method. Am J.ClinPathol. 45, 493-496.

Camus, A., Berliner, A., Clauss, T., Hatcher, N., Marancik, D., 2013. *Serratiamarcescens* infection in a bonnethead shark. John Wiley & Sons Ltd.

Chovatiya, S., Ingale, S., Rana, D., Patel, P., 2014. Isolation, Characterization and Optimization of Amylase Producing Microorganism from Gastrointestinal Tract of *Catla catla*. International Journal of Innovative Research in Science, Engineering and Technology. 3, 112-118.

Ellis, F., 2001. The livelihoods approach and management of small-scale fisheries. Marine Policy. 25, 377-388.

Food and Agriculture Organization of The United Nations (FAO). The State of Food and Agriculture 2013. Innovative food systems. Rome, Italy.

Kabata, Z. 2008.Parasites and Diseases Tropics. Taylor publ.london.92pp

Kurz, C. L., Chauvet, S., Andres, E., Aurouze, M., Vallet, I., Michel G. P. F., Uh, M., Celli, J., Filloux, A., de Bentzmann, S., Steinmetz, I., Hoffman J. A., Finlay, B. B., Gorvel, J. P., Ferrandon, D., Ewbank, J. J., 2003. Virulence factors of the human opportunistic pathogen *Serratiamarcescens* identified by in vivo screening. The EMBO Journal 22, 1451–1460.

Nabeel M Alikunhi., Zenon B Batang., Haitham A AlJahdali., Mohammed A.M. Aziz and Abdulaziz M Al-Suwailem.,2016. Culture-dependent bacteria in commercial fishes: Qualitative assessment and molecular identification using 16S rRNA gene sequencing., Saudi Journal of Biological Sciences (In Press).

Obi, S.K.C., and Krakowiaka,A.,1983. Theory and practices of food Microbiology.(Unpublished manual)

Omprakash, D., and Ravi, D. B., 2015. *Pseudomonas fluorescens* associated with Bacterial Disease in *Catla catla* in Marathwada Region of Maharashtra. IJBR. ISSN 0976-2612, Online ISSN 2278-599X, 6, 189-195.

Poornima, K., 2012. Enhanced Production of Prodigiosin Production in *Serratia marcescens*. Journal of Applied Pharmaceutical Science.02, 138-140.

Saidenberg, A.B., Teixeira, R. H., Astolfi-Ferrieira, C. S., Kn€obl, T., Ferreira, A. J., 2007. *Serratiamarcescens* infection in a swallow-tailed hummingbird. Journal of Wildlife Disease. 43, 107-110.

Shahitha, S., 2012. Enhanced Production of Prodigiosin Production in *Serratia marcescens*. Journal of Applied Pharmaceutical Science. 02, 138-140.

Sharma, M., Shrivastav, A. B., Sahni, Y. P., Pandey, G., 2012. Overviews of the treatment and control of common fish diseases, Inter. Res. J. Pharma. 3, 123-127.

Subhash ,W.,Shivaji, K., Navnath, P., Shivaji ,C.,2015. Opportunistic Pathogens in the Mucous of Skin, Gills, Fins and Mouth of Labeo rohita. International

Journal of Fisheries and Aquatic Studies 3(1), 169-172

Sujatha, K., MacKinnnon, S. L., Ross, N. W., 2011. Isolation of human pathogenic bacteria in two edible fishes, *Priacanthushamrur* and *Megalaspiscordyla* at Royapuram waters of Chennai, India. Indian Journal of Science and Technology. **4**: 539-541.

Sujatha, M., Dhasarathan, P., Nivetha, V., Kuberan, T., 2013. Pathogenicity of bacterial isolates to *Catla catla*.